A quote that presents landslide mitigation from a general perspective

To quote Highland an Bobrowsky (2008): "Vulnerability to landslide hazards is a function of a site's location (topography, geology, drainage), type of activity, and frequency of past landslides. The effects of landslides on people and structures can be lessened by total avoidance of landslide hazard areas or by restricting, prohibiting, or imposing conditions on hazard-zone activity. Local governments can accomplish this through land use policies and regulations. Individuals can reduce their exposure to hazards by educating themselves on the past hazard history of a desired site and by making inquiries to planning and engineering departments of local governments. They could also hire the professional services of a geotechnical engineer, a civil engineer, or an engineering geologist who can properly evaluate the hazard potential of a site, built or unbuilt."

3.7 Highlight on CROM DSS

The Catchment Restoration Opportunity Mapping Support System (CROM DSS) developed by W4G for the IWRM programme, a nationally-adopted tool for the mapping of soil erosion risk areas in Rwanda, proposes several land use options to mitigate the degradation of soil erosion by rainfall (e.g. agroforestry, progressive and bench terraces, forestation). This document is framed around a RUSLE soil erosion analysis, i.e. a physically plausible empirical method for predicting soil erosion (Renard et al., 1997). The RUSLE-based modelling approach provides estimates of the potential rates of soil displacement by water erosion (soil erosion). More specifically, the RUSLE modelling only predicts soil losses caused by sheet and rill erosion. In short this model does not handle at all other soil erosion processes such as gullying. In addition, gravity-driven processes such as landslides are not at all seized by this model. Therefore, what is suggested to managed soil erosion in CROM DSS must be considered with the highest care when mitigating the landslides. Indeed, reducing soil erosion, means reducing the runoff water. If the runoff is reduced, this means that the infiltration of the water is increased. If the infiltration is increased, this means that the soil/regolith water content is increased, hence creating conditions that are more favourable for landsliding (Sidle and Ochiai, 2006; Lu and Godt, 2013, Table 13, Figure 13).

A very striking feature in soil water conservation is that of agricultural terraces (Figure 16 – Figure Appendix 3.1); i.e. soil conservation practices that are commonly adapted in many regions of Rwanda e.g. the Ministry of Agriculture and Animal Resources (MINAGRI) set up the fourth Strategic Plan Agriculture Transformation for 2018 – 2024 where bench terraces are highlighted as a mean to fight against soil erosion).